

QB 04: Mathematical Foundations for MCA — Question Bank with Answer Keys

Course: Mathematical Foundations for Computer Applications (Discrete Math, Linear Algebra & Probability)

Target: VTU MCA Semester 1 (Tailored for students who want intuitive, step-by-step clarity)

Scheme: VTU 2022 / 2024 Scheme

Contents: Logic Truth Tables, Equivalence Relations, Graph Handshaking, Matrix Eigenvalues, Bayes' Theorem, 15 MCQs with Explanations

Syllabus Outline (VTU 5 Modules)

- **Module 1:** Propositional Logic, Truth Tables, Logical Connectives, Tautology, De Morgan's Laws, Predicates, and Quantifiers ($\forall$, $\exists$).
- **Module 2:** Set Theory, Relations (Reflexive, Symmetric, Transitive, Equivalence Relations, Equivalence Classes), Posets & Hasse Diagrams, Functions (One-to-one, Onto, Invertible).
- **Module 3:** Graph Theory: Basic Terminology, Handshaking Theorem, Regular/Bipartite Graphs, Eulerian & Hamiltonian Graphs, Adjacency Matrices, Planar Graphs & Euler's Formula.
- **Module 4:** Linear Algebra: Matrix Operations, Determinants, Matrix Inverse, Systems of Linear Equations (Cramer's Rule, Gaussian Elimination), Eigenvalues & Eigenvectors.
- **Module 5:** Probability & Statistics: Axioms of Probability, Conditional Probability, Bayes' Theorem, Discrete Random Variables, Expectation (Mean) & Variance.

SECTION 1: High-Yield MCQs with Answer Key

Q1. A compound proposition that is always TRUE regardless of the truth values of its individual variables is called a:

- A) Contradiction
- B) Contingency
- C) Tautology
- D) Fallacy

Answer: C

Explanation: A tautology evaluates to True (T) across every single row of its truth table (e.g., $p \vee \neg p$).

Q2. According to De Morgan's Law, what is the logical negation of $(p \wedge q)$?

- A) $\neg p \wedge \neg q$
- B) $\neg p \vee \neg q$
- C) $p \vee q$
- D) $\neg p \rightarrow q$

Answer: B

Explanation: $\neg(p \wedge q) \equiv \neg p \vee \neg q$. Negating an "AND" statement flips it into an "OR" of the negated variables.

Q3. A relation R on set A is an "Equivalence Relation" if it is:

- A) Reflexive, Antisymmetric, Transitive
- B) Reflexive, Symmetric, Transitive
- C) Irreflexive, Symmetric, Transitive
- D) Symmetric and Transitive only

Answer: B

Explanation: An equivalence relation must satisfy three properties: (1) Reflexive, (2) Symmetric, and (3) Transitive. (If antisymmetric instead of symmetric, it is a Partial Order / Poset).

Q4. The Handshaking Lemma states that in any undirected graph $G = (V, E)$:

- A) $\sum_{v \in V} \text{deg}(v) = |E|$
- B) $\sum_{v \in V} \text{deg}(v) = 2|E|$
- C) $\sum_{v \in V} \text{deg}(v) = |V| \times |E|$
- D) Total odd vertices is always odd

Answer: B

Explanation: Every edge contributes 1 to the degree of each of its two endpoints; thus, the sum of all vertex degrees is always twice the number of edges ($2|E|$).

Q5. If an undirected graph has 10 edges, what is the sum of the degrees of all vertices?

- A) 10
- B) 15
- C) 20
- D) 25

Answer: C

Explanation: By Handshaking Lemma: $\sum \text{deg}(v) = 2 \times |E| = 2 \times 10 = 20$.

Q6. For a connected planar graph with V vertices, E edges, and F faces/regions, Euler's formula states:

- A) $V - E + F = 2$
- B) $V + E - F = 2$
- C) $V - E - F = 0$
- D) $V \times F = E$

Answer: A

Explanation: Euler's Planar Formula: $V - E + F = 2$ for any connected planar graph.

Q7. If the determinant of a square matrix A is zero ($\det(A) = 0$), the matrix is called:

- A) Orthogonal
- B) Non-singular
- C) Singular (and has no inverse)
- D) Symmetric

Answer: C

Explanation: A matrix with determinant 0 is singular and cannot be inverted because $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$, which would require division by zero.

Q8. The eigenvalues (λ) of a square matrix A are roots of which characteristic equation?

- A) $\det(A) = 0$
- B) $\det(A - \lambda I) = 0$
- C) $A \cdot \lambda = I$
- D) $\text{Tr}(A) - \lambda = 0$

Answer: B

Explanation: Non-trivial solutions to $Ax = \lambda x$ implies $(A - \lambda I)x = 0$ require that the determinant $|A - \lambda I| = 0$.

Q9. If events A and B are mutually exclusive, what is $P(A \cap B)$?

- A) 1
- B) $P(A) \times P(B)$
- C) 0
- D) $P(A) + P(B)$

Answer: C

Explanation: Mutually exclusive events cannot happen at the same time; their intersection is an empty set $\emptyset$, so $P(A \cap B) = 0$.

Q10. Bayes' Theorem computes the posterior probability $P(A|B)$ as:

- A) $\frac{P(B|A) P(A)}{P(B)}$
- B) $\frac{P(A \cap B)}{P(A)}$
- C) $P(A) + P(B|A)$
- D) $P(A) \times P(B)$

Answer: A**Explanation:** Bayes' formula is $P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$. **SECTION 2: Step-by-Step Concept Questions (4–6 Marks)****Q11. Define Equivalence Relation. Prove that congruence modulo n on integers ($a \equiv b \pmod{n}$) is an equivalence relation.****Answer:****Definition:** A relation R on a set A is an **equivalence relation** if it satisfies:

1. **Reflexive:** $\forall a \in A, (a, a) \in R$.
2. **Symmetric:** If $(a, b) \in R$ implies $(b, a) \in R$.
3. **Transitive:** If $(a, b) \in R$ and $(b, c) \in R$ implies $(a, c) \in R$.

Proof for Congruence Modulo n :Relation: $a \equiv b \pmod{n}$ iff $(a - b)$ is divisible by n (i.e., $a - b = k \cdot n$ for some integer k).**1. Reflexive:**

- $a - a = 0 = 0 \cdot n$.
- Since 0 is divisible by n , $a \equiv a \pmod{n}$. $\implies$ **Reflexive**.

2. Symmetric:

- Assume $a \equiv b \pmod{n}$. Then $a - b = k \cdot n$.
- Multiply both sides by -1 : $b - a = (-k) \cdot n$.
- Since $-k$ is also an integer, $(b - a)$ is divisible by n .
- Hence, $b \equiv a \pmod{n}$. $\implies$ **Symmetric**.

3. Transitive:

- Assume $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$.
- Then $a - b = k_1 \cdot n$ and $b - c = k_2 \cdot n$.
- Adding both equations: $(a - b) + (b - c) = a - c = (k_1 + k_2) \cdot n$
- Since $(k_1 + k_2)$ is an integer, $(a - c)$ is divisible by n .
- Thus, $a \equiv c \pmod{n}$. $\implies$ **Transitive**.

Conclusion: Since the relation is reflexive, symmetric, and transitive, it is an **Equivalence Relation**. $\blacksquare$

Q12. State the Handshaking Theorem. Prove that in any graph, the number of vertices with odd degree is always even.

Answer:

Handshaking Theorem:

In any undirected graph $G = (V, E)$: $\sum_{v \in V} \text{deg}(v) = 2|E|$

Proof: Divide the vertex set V into two disjoint subsets:

- V_1 : Set of vertices with **even** degree.
- V_2 : Set of vertices with **odd** degree.

Then: $\sum_{v \in V} \text{deg}(v) = \sum_{v \in V_1} \text{deg}(v) + \sum_{v \in V_2} \text{deg}(v) = 2|E|$

1. Notice that $2|E|$ is always an **even number**.
2. The sum of degrees of vertices in V_1 ($\sum_{v \in V_1} \text{deg}(v)$) is a sum of even numbers, which is always **even**.
3. Subtracting the even sum: $\sum_{v \in V_2} \text{deg}(v) = 2|E| - \sum_{v \in V_1} \text{deg}(v) = \text{Even} - \text{Even} = \text{Even}$
4. For the sum of odd numbers to be **even**, there must be an **even number of terms** in the sum!
5. Therefore, $|V_2|$ (the count of vertices with odd degree) must be **even**. $\square$



SECTION 3: VTU Model Numerical Problems (10–12 Marks)

Q13. [VTU Model QP - Truth Table Proof]

Construct a Truth Table to prove that the following proposition is a Tautology:

$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

(Law of Hypothetical Syllogism)

Answer:

Let us evaluate step-by-step for all $2^3 = 8$ truth assignments of (p, q, r) :

- Recall: An implication $A \rightarrow B$ is False **only when $A = \text{True}$ and $B = \text{False}$** ; in all other cases, it is True (True).

p	q	r	$p \rightarrow q$	$q \rightarrow r$	Premise: $(p \rightarrow q) \wedge (q \rightarrow r)$	Conclusion: $p \rightarrow r$	Full Expression: $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	T	T
T	F	T	F	T	F	T	T
T	F	F	F	T	F	T	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

Conclusion:

Every single entry in the final column is **True (True)**.

Therefore, the proposition is a **Tautology**. $\square$

Q14. [VTU Model QP - Linear Algebra: Eigenvalues & Eigenvectors]

Find the Eigenvalues and corresponding Eigenvectors for the matrix:

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

Answer:

Step 1: Characteristic Equation

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{vmatrix} = 0$$

$$(4 - \lambda)(3 - \lambda) - (1 \times 2) = 0 \implies 12 - 4\lambda - 3\lambda + \lambda^2 - 2 = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

Factorizing the quadratic equation: $(\lambda - 5)(\lambda - 2) = 0 \implies \lambda_1 = 5, \lambda_2 = 2$ The **Eigenvalues** are **5** and **2**.

Step 2: Eigenvector for $\lambda_1 = 5$

$$\text{Solve } (A - 5I)X = 0: \begin{pmatrix} 4 - 5 & 1 \\ 2 & 3 - 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\text{From row 1: } -x_1 + x_2 = 0 \implies x_1 = x_2.$$

$$\text{Let } x_2 = 1 \implies x_1 = 1.$$

$$\text{Eigenvector for } \lambda = 5: X_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Step 3: Eigenvector for $\lambda_2 = 2$

$$\text{Solve } (A - 2I)X = 0: \begin{pmatrix} 4 - 2 & 1 \\ 2 & 3 - 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\text{From row 1: } 2x_1 + x_2 = 0 \implies x_2 = -2x_1.$$

$$\text{Let } x_1 = 1 \implies x_2 = -2.$$

$$\text{Eigenvector for } \lambda = 2: X_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

Q15. [VTU Model QP - Bayes' Theorem Problem]

In a software company, a spam filter evaluates incoming emails. It is known that 20% of all emails are spam (\$S\$) and 80% are legitimate/ham (\$H\$).

- **If an email is spam, the probability that it contains the word "Discount" is 90% ($P(D|S) = 0.90$).**
- **If an email is legitimate, the probability that it contains the word "Discount" is only 5% ($P(D|H) = 0.05$).**

If a randomly received email contains the word "Discount", what is the probability that it is actually SPAM?

Answer:

Step 1: Identify Given Probabilities

- Prior probability of Spam: $P(S) = 0.20$
- Prior probability of Legitimate (Ham): $P(H) = 0.80$
- Likelihood of "Discount" given Spam: $P(D|S) = 0.90$
- Likelihood of "Discount" given Ham: $P(D|H) = 0.05$

Step 2: Total Probability of an email containing "Discount" ($P(D)$)

By the Law of Total Probability: $P(D) = P(D|S) \cdot P(S) + P(D|H) \cdot P(H)$
 $P(D) = (0.90 \times 0.20) + (0.05 \times 0.80)$
 $P(D) = 0.18 + 0.04 = \mathbf{0.22}$

Step 3: Apply Bayes' Theorem

We want to find $P(S|D)$ (the probability that the email is Spam, given that it contains "Discount"): $P(S|D) = \frac{P(D|S) \cdot P(S)}{P(D)}$

$$P(S|D) = \frac{0.90 \times 0.20}{0.22} = \frac{0.18}{0.22} = \frac{18}{22} = \frac{9}{11} \approx \mathbf{0.8182}$$

Final Answer:

The probability that the email is spam is $\mathbf{81.82\%}$.

 Key Takeaway Checklist for Exam Day

- [] For truth tables, double-check that $p \rightarrow q$ is false **ONLY** when p is True and q is False.
- [] For graph problems, always verify using the Handshaking formula: $2 \times \text{Edges} = \text{Sum of degrees}$.
- [] When finding eigenvalues, confirm: $\lambda_1 + \lambda_2 = \text{Trace}(A)$ (sum of diagonal entries) and $\lambda_1 \times \lambda_2 = \det(A)$.
 (Check for matrix A : $5 + 2 = 7 = 4 + 3$, and $5 \times 2 = 10 = (12 - 2)$. Matches perfectly!).